

INTRODUCTION

Traffic signals play a crucial role in managing vehicular and pedestrian movement at road intersections. Automated traffic light controllers are widely used across the world to reduce human error, maintain order, and enhance safety. This project replicates the working of a basic three-light traffic signal using simple electronic components. It eliminates the need for programming or microcontrollers and instead uses basic digital logic circuits.

The main ICs used in the project are the 555 Timer, which generates time-based pulses, and the CD4017 Decade Counter, which uses those pulses to activate its outputs in a sequential manner. Each output corresponds to a traffic light:

- Red (Stop)
- Yellow (Ready)
- Green (Go)

The entire system operates from a 9V battery, but digital ICs require just 5V, so a 7805 voltage regulator is used to convert 9V to a stable 5V DC. Large and small capacitors are used for smoothing and noise filtering, ensuring stable operation of the ICs.

This project is ideal for understanding the real-world application of sequential circuits and timing mechanisms in everyday systems such as traffic management.

OBJECTIVE

The main objectives of this project are:

1. To design a simple and automatic three-light traffic controller.
2. To use the 555 Timer IC in astable mode to generate clock pulses.
3. To use the CD4017 Decade Counter IC to decode pulses into sequential outputs.
4. To demonstrate the use of voltage regulators for stable DC supply.
5. To understand timing circuits, counters, and LED interfacing.
6. To construct the circuit on a breadboard for practical learning.

BLOCK DIAGRAM

9V Battery → 7805 Regulator → 5V Output → 555 Timer IC → CD4017 Counter IC → LEDs (Red,Yellow,Green)

COMPONENTS REQUIRED

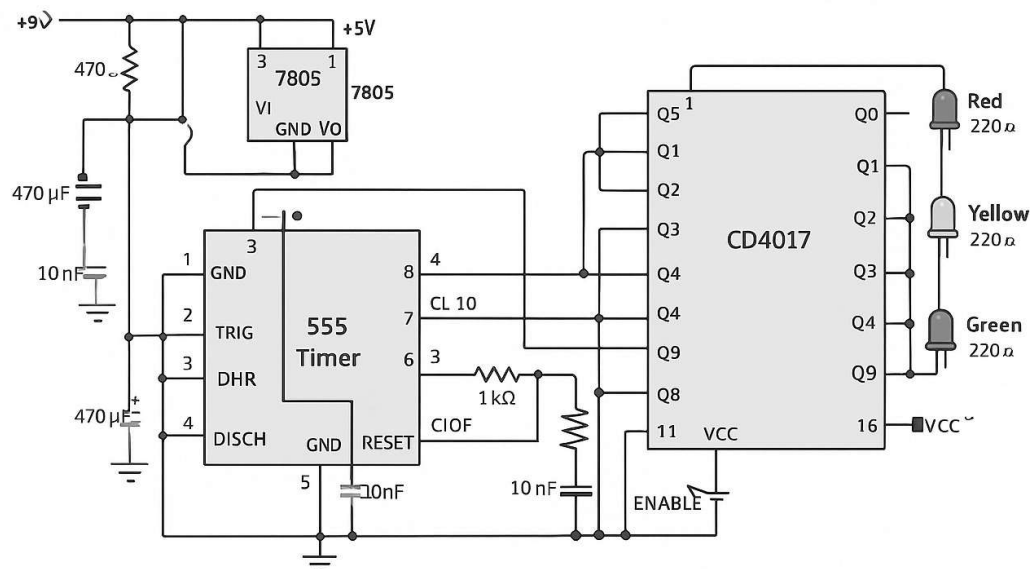
1. **555 Timer IC**
2. **CD4017 Decade Counter IC**
3. **LEDs**
 - Red LED
 - Yellow LED
 - Green LED
4. **Resistors**
 - $3 \times 220 \Omega$ (for LEDs)
 - $1 \times 220 \Omega$ (timing resistor)
 - $1 \times 1 \text{ k}\Omega$ (timing resistor)
5. **Capacitors**
 - $470 \mu\text{F}$ electrolytic capacitor
 - 10 nF ceramic capacitor
6. **7805 Voltage Regulator (5V output)**
7. **9V Battery or 9V Power Supply**
8. **Breadboard**
9. **Connecting Wires**

CIRCUIT DIAGRAM

DESCRIPTION :

Main Connections

- Pin 3 of the 555 timer IC is connected to pin 14 (clock input) of CD4017.
- Output of the 7805 voltage regulator (5V) is connected to the power supply pins of both ICs (555 and CD4017).
- LEDs are connected to the outputs Q0, Q1, and Q2 of the CD4017 IC.
- Capacitors are connected across the voltage regulator and the timer IC for filtering and stabilization.



WORKING PRINCIPLE

1. Power Supply Section

The circuit uses a 9V battery as the main power source.

The 7805 Voltage Regulator steps down 9V to a stable 5V DC, which is required for the 555 and CD4017 ICs.

- The 470 μ F capacitor smooths the input voltage.
- The 10 nF capacitor reduces high-frequency noise.

2. 555 Timer in Astable Mode

The 555 Timer IC generates continuous pulses.

Pin connections:

- Pin 1 $\rightarrow$ Ground
- Pin 8 $\rightarrow$ +5V
- Pin 4 $\rightarrow$ +5V (Reset disabled)
- Pin 5 $\rightarrow$ 10 nF capacitor to GND
- Pin 2 & Pin 6 shorted
- R1 (1 k Ω) between Pin 7 and +5V
- R2 (220 Ω) between Pin 7 and Pin 2
- C1 (470 μ F) between Pin 2/6 and GND
- Pin 3 = Output clock

This output goes to the CD4017 and controls the timing of lights.

3. CD4017 Decade Counter

Pin connections:

- Pin 16 → +5V
- Pin 8 → Ground
- Pin 14 → Clock input (from 555 Pin 3)
- Pin 13 → Ground (enable counting)
- Pin 15 → Reset (connected to Q3 or Q4 depending on cycle length)

Sequential outputs:

- Q0 (Pin 3) → Red LED
- Q1 (Pin 2) → Yellow LED
- Q2 (Pin 4) → Green LED

After Q2, RESET brings the counter back to Q0.

4. LED Section

Each LED has a 220 Ω resistor on the positive terminal.
Connections:

- CD4017 Pin 3 → 220 Ω → Red LED → GND
- CD4017 Pin 2 → 220 Ω → Yellow LED → GND
- CD4017 Pin 4 → 220 Ω → Green LED → GND

5. How the Sequence Works

1. 555 sends a pulse → CD4017 output moves to next pin
2. First pulse → Q0 → Red LED glows
3. Second pulse → Q1 → Yellow LED glows
4. Third pulse → Q2 → Green LED glows
5. Fourth pulse triggers RESET → back to Red

APPLICATIONS

- Demonstration in schools and colleges
- Traffic model exhibitions
- Understanding sequential logic
- Basic automation systems
- LED sequencing projects

ADVANTAGES

- No programming required
- Low power consumption
- Easy to modify timing
- Very low cost
- Beginner friendly

CONCLUSION

The Traffic Light Controller using 555 Timer IC and CD4017 IC is an excellent project to understand digital electronics, timing circuits, and sequential control. The combination of a 555 timer and 4017 counter provides a simple and reliable traffic light sequence without using any microcontroller.

The project successfully demonstrates the working of Red, Yellow, and Green lights using basic components, a regulated 5V power supply, and timing capacitors.

This system is stable, cost-effective, and ideal for educational purposes and practical demonstrations.

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